CS579 Final Project Report: Ridership Prediction for Expanding Metro Networks

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Introduction

This is the project report for the final project of *CS-579, Online Social Network Analysis*. This report begins by introducing the problem statement and discussing the motivation for selecting it. It then talks about the design of the project, along with all the assumptions and simplifications made as part of the design decisions. It then proceeds to talk about the data sources and the data used in the project, along with data cleaning and pre-processing, followed by implementation of the project. After that, the report discusses the results and future steps. Finally, the reference section gives information about all the sources, such as research papers, data portals, etc, that were used for the project. It also contains a section mentioning all the AI help that was taken to implement this project.

1 Problem Statement & Motivation

The goal of this project is to use machine learning techniques (Neural Networks) and concepts of graphs to predict the ridership of a metro/subway/L station, specifically, the newly created and/or updated stations. A station is considered updated in this context when there is a change in the number of lines/trains passing through it. This project was implemented for the 'L' trains in the city of Chicago.
The motivation(s) behind doing this project were:

- Expansion of metro networks is an ongoing and continuous project throughout the world, and it is a resource and money-intensive endeavour. Having a prediction algorithm that can give an idea of potential ridership can help cities better plan their metro expansion efforts.
- For the final project, I wanted to continue working on the city of Chicago. However, I wanted to move beyond the community regions and neighbourhoods and focus on public transit in the city of Chicago.

This project is an implementation of an existing research paper: F. Ding, Y. Liang, Y. Wang, Y. Tang, Y. Zhou, and Z. Zhao, "A graph deep learning model for station ridership prediction in expanding metro networks," in Proc. 2nd ACM SIGSPATIAL Int. Workshop Advances Urban-AI, 2024, pp. 6–14, doi:10.1145/3681780.3697247.
The authors of the paper implement a graph attention network-based ridership prediction for expansions done in the metro network of Shanghai, China. This project very closely follows the paper, and a subset of techniques and data was used/implemented from Ding et al. due to data unavailability and/or compute-related blockers.

2 Project Design and Data

The project mixes graphs and neural networks. Graphs are used to collect and create features, and then a small neural network is trained on these features to generate ridership predictions. To be specific, the prediction is the total monthly ridership (both the number of people getting on trains from the station and people getting off the station).

Figure 1: Data Sources

2.1 Data

Data from different sources and/or fields is used to create the training dataset. The following categories of data were used in this project:

- **Per station ridership data:** Monthly ridership for each station serves as the label for prediction in training the neural networks. Ridership for each station for every month it was operable within a certain time period is required. The ridership data was obtained from Chicago Data Portal. The data was in the form of a CSV file, with average weekday, average Saturday, average Sunday/holiday rides and the total monthly ridership for each station in the CTA L network starting from January 2001 to September 2025. Out of this data, monthly ridership starting from January 2020 to September 2025 (the last 5 years) was used.

- **Bus Stops Data:** Data about the location of bus stops and bus lines are used to create multimodal features for model training. For the city of Chicago, coordinates of all the bus stops, and the bus routes/lines they belong to are obtained from Chicago Data Portal. This data is a shapefile containing coordinates for all the bus stops in the city of Chicago, their names and the routes/lines they serve.

- **Population Density & Housing Value Data:** The median population and housing value data are used to create what is called the built environment data. These datasets are collected from the American Community Survey datasets. Data for all the block groups Illinois was taken for the year 2023 and was then filtered for only the block groups in the city of Chicago. To filter, the boundary of the city was taken from the Chicago Data Portal, and only the block groups that were inside these boundaries were taken.

- **'L' Stations and Lines/Tracks Shapefiles:** These shapefiles are used to create the graph for the entire 'L' networks. Additionally, the shapefile for the stations will also be used to create different features. These shapefiles were downloaded from Chicago Data Portal. These can be obtained upon searching for CTA 'L' (Rail) Stations and CTA 'L' (Rail) Lines, respectively.

2.2 Project Design

The project design consists of two parts:

1. Graph Creation & Feature Engineering/Creation

2. Neural Network Architecture

2.2.1 Graph Creation and Feature Engineering

The entire network of the 'L' was created as a graph, with each station being a node with an edge connecting the nodes if the two stations are connected by a line. Note that if two stations are connected via a track but there does not exist a line that connects those two stations, the graph does not connect the corresponding edges. Since, to the best of my knowledge, there does not exist an adjacency matrix for creating this 'L' network. To create the graph, for all the rails from the rails shapefile, the closest stations to each end were obtained, and an edge between the two was created in the graph. This gave a fairly accurate graph, the mistakes were then manually corrected to obtain the final graph, which was the same as the 'L' network.

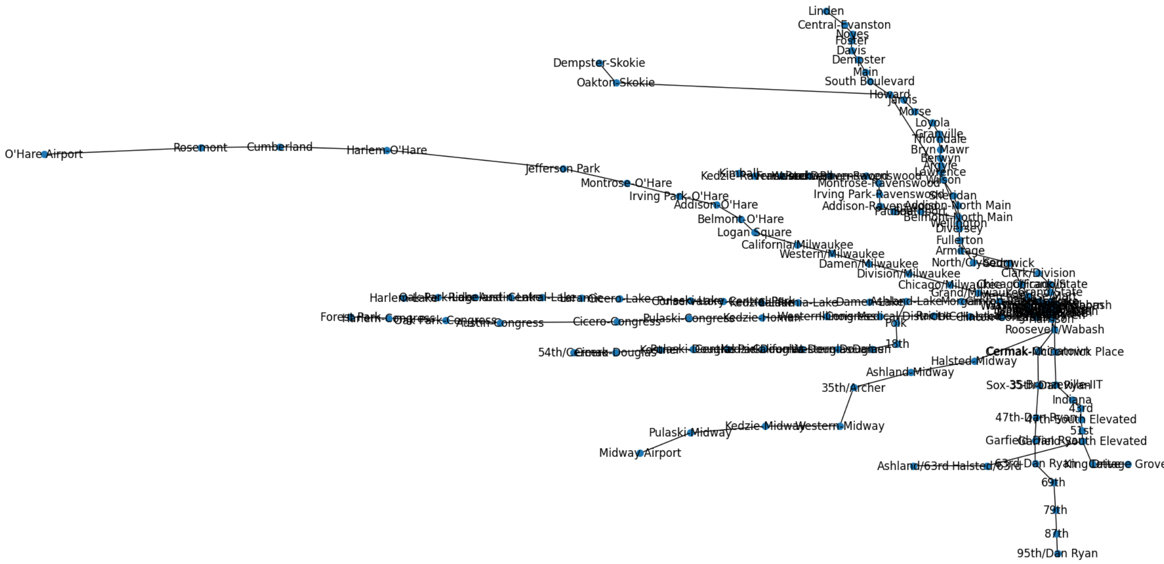


Figure 2: Graph of the 'L' Network

A note on the temporal nature of the graph: Like other metro/subway networks, the 'L' network also changes over time. Any updates, such as shutting stations down, opening new stations and line expansions, change the network and thus also the graph. Some features in the training data for this project respect and contain these temporal changes. To do this, multiple graphs were made for every time the network changed. These dedicated 'time-sensitive' sets of graphs were then used to create certain temporal features. Between 2020 to the present, the 'L' network changed 5 times:

1. June 2021: Berwyn and Lawrence red line stations were closed as part of the Red-Purple line modernisation project.
2. August 2023: Racine blue line station was shut down for station repairs and rebuilding.
3. September 2023: Racine blue line station was added back
4. August 2024: The new Damen green line station was opened.
5. June 2025: The Berwyn and Lawrence Red Line stations were reopened.

For the features, three different categories of features were created:

1. **Network Structure Features:** These are graph-based features such as centralities, degrees, etc. The idea behind collecting these features is to quantify the *centrality* and *centrality-derived importance* of each station, as more *central* features tend to be the bigger interchange stations with higher passenger traffic. At the same time, whether a station is the terminal station for a line was also recorded. For each node/station, this feature is a vector of 9 elements:

   [Degree Centrality, Closeness Centrality, Betweenness Centrality, Eigenvector Centrality, Katz Centrality, Degree of Node, Is node an interchange?, Is node a terminal station?, No. of lines passing through the node/station]

   These features are temporal, meaning their values change as the 'L' network changes. Due to the 5 changes mentioned above, 6 graphs (one for before June 2021, where every station except the new Damen Green Line station was present) were created, and features were calculated for each node for each of these graphs.

Finally, during concatenation, to create the training data, the structure features for the correct time period are taken.

2. **Built Environment Features:** These features represent/give an idea about the population density and infrastructure, and how built up the area around a station is in general. The idea is, the more populous and built up the area around a station is, its likely that more people will use the trains, compared to an area with relatively less population density. To do this, the population density and housing value within a 500 m radius around each station are collected. To collect these features, the median population and housing value for each block group whose centroid is within this 500 m radius were collected and averaged to get the final value. Since the data was collected from ACS, which gives a measure of error for each estimate, in addition to the estimate averages, the measure of error averages was also collected. This finally gave, for each station/node, a vector of length 4, [population average, population measure of error, housing value average, housing value measure of error].

3. **Multimodal Features:** These features contain information about how many other modes of transit are active around a station. Different modes of transit feed into and from each other, for eg, local buses can take people to the subway/metro and vice versa. Thus, having a number of different transit modes active around a station can also increase the ridership. These are also calculated in a radius of 500 m around the stations. For this, the number of bus stops and bus lines is obtained for each station and packed as a vector of two elements.

2.2.2 Neural Network Architecture

To make ridership predictions, a small fully connected network is used as the predictor. It is trained on the concatenated features described above and is trained to predict ridership. The architecture of the predictor is as follows:

- Number of layers: 3
- Number of units in each layer: 128, 64, 32
- Activation Function: ReLU for the hidden layers, linear activation for the output.

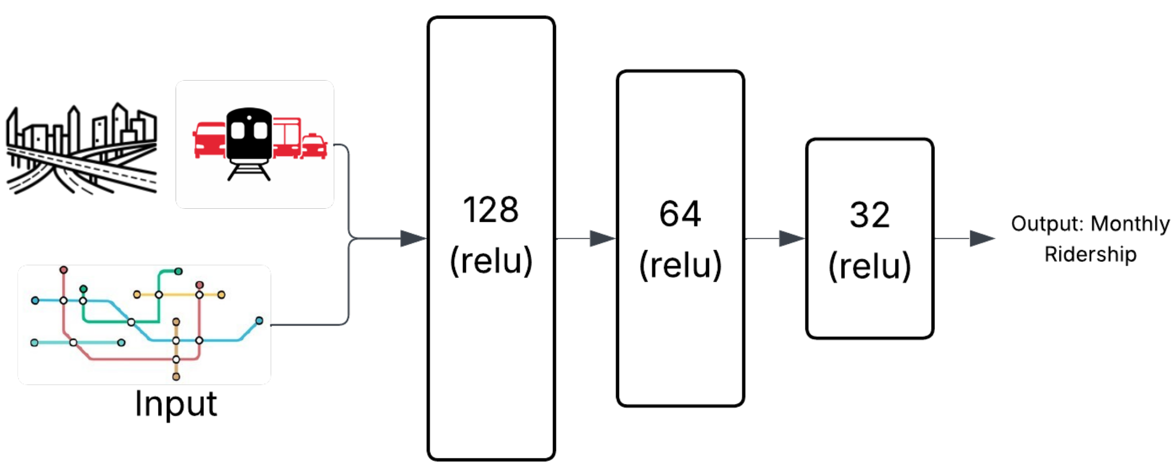


Figure 3: Predictor Architecture

2.2.3 Assumptions & Simplifications:

The following assumptions were made as part of the project:

- The shapefiles for stations, bus stops and block groups are up to date.
- All the stations fall inside the city boundaries obtained from the Chicago data portal.
- The naming of some stations is different from that given in the CTA map, for example, in the shapefiles, and the ridership counts, certain Red Line stations, such as Jackson, are referred to as Jackson/State and the Jackson Blue Line station is referred to as Jackson/Milwaukee. A manual check was done on these stations to find out which is which based on the adjacency as well as the branch names from Chicago"L".org. An assumption is made that these *guesses* are correct.
- There are some stations that are connected via walkways, for example Library and the Jackson Red and Blue line stations. Since they're not connected to each other via tracks, and exist as separate stations, they are not connected via an edge. An edge only exists between stations if they're connected by tracks, and not walkways.

The following simplifications were made as part of the project:

- The ACS data has a measure of error, and high error estimates need to be handled. However, in the interest of time, this is not done, and all estimates are used. Instead of processing, the measure of error is also introduced as a feature in the model, *hoping* that the model learns from it.
- Since block groups can have a large variety when it comes to their geometry, to check which block group lies within the 500 m radius, the centroid for each is calculated, and if the centroid lies inside the radius, the entire block group is assumed to lie inside the radius.

3 Project Execution: Training

3.1 Training Data Preparation

Once the data collection is done training dataset is created by concatenating all the features together. To be more specific, from the ridership dataset, since each row contains the monthly ridership for a station for a specific month, the 3 feature vectors for that station are concatenated together, and the ridership is concatenated to this final vector. Since the network structure features are temporal, the features from the graph of the period are taken, to which the month-year in the ridership dataset row belongs.

3.1.1 Data Cleaning and Processing

First, the final dataset is checked to ensure that no null or unexpected values, such as strings, etc., are not present. After that, all the features, except the ridership, are normalised to a [0-1] scale. Finally, the model was split into training and testing sets with an 80-20 train-test ratio.

3.2 Model Training

The model was trained for *1000 epochs*, with a *learning rate of 0.001*. The model was trained to minimise the *RMSE loss*, which also served as the performance metric for the model. The training plot is given below:

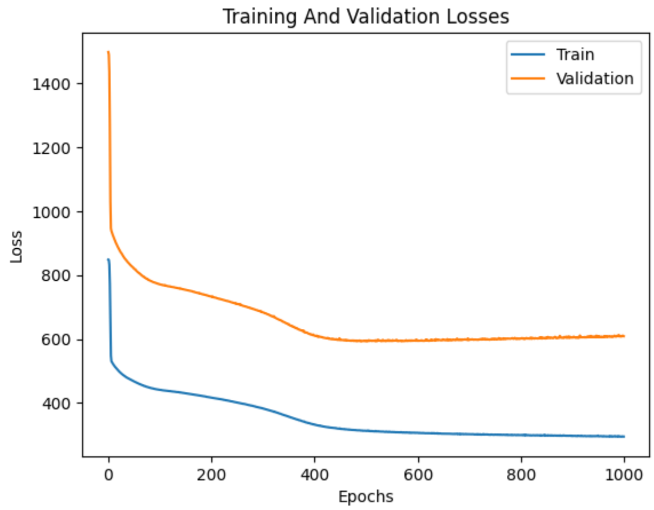


Figure 4: Model Training Plot

Discussion of training results

From the training plot, it can be seen that there is a fairly large difference between the train and validation(test) RMSE, and it seems that the model is overfitting. However, this was the best performing model that could be obtained on this data, and the prediction of the model on the test data was not very off, and therefore, this model was chosen for further analysis.

4 Results & Discussion

After the model was trained, monthly ridership predictions were made for the following three stations:

4.1 Results

1. Damen: Green Line
2. Berwyn: Red Line
3. Lawrence: Red Line

4.1.1 Results: Damen- Green Line

The Damen station on the Green Line is a newly constructed and opened station in August 2024 between California and Ashland stations. Predictions for this station are as follows:

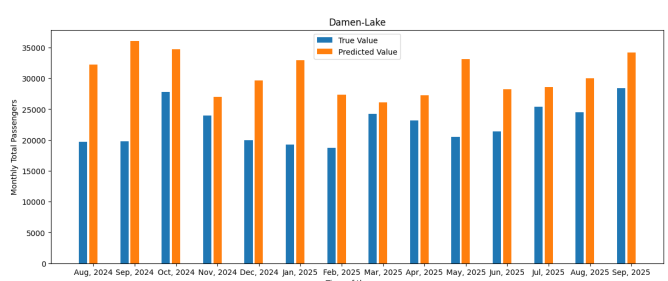


Figure 5: Predictions: Damen

From the graph, it can be seen that the predictions are not too bad. Some of them come fairly close to the original value, such as Nov 24, Mar 25, Apr 25, etc, while others, such as Aug and Sep 24, are very off.

4.1.2 Results: Berwyn- Red Line

The Berwyn station was operational till June 2021, after which it was closed for reconstruction as part of the Red-Purple line modernisation project. Predictions for this station are given below:

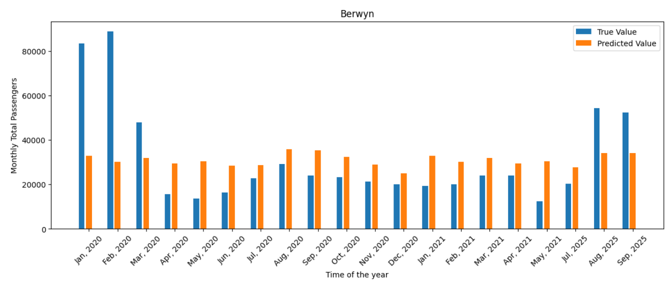


Figure 6: Predictions: Berwyn

For this station, a similar story to that in Damen can be seen. However, for the first two months, Jan 2020 and Feb 2020, the model severely underestimates the ridership. While this project is about prediction for newly opened/reopened/updated stations, predictions for this station are made for the time it was already active as well, mostly because data for only 3 months was available to compare after its reopening in 2025.

4.1.3 Results: Lawrence- Red Line

The Lawrence station was operational till June 2021, after which it was closed for reconstruction as part of the Red-Purple line modernisation project. Predictions for this station are given below:

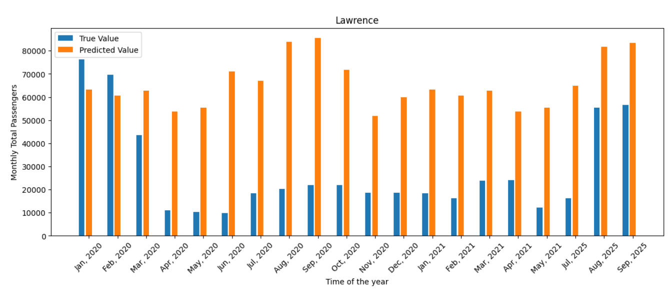


Figure 7: Predictions: Lawrence

Unlike Berwyn and Damen, the predictions for this station are largely very inaccurate, with the model overestimating the ridership a lot. With the exception of Jan 2020 and Feb 2020, where it approaches the true number. This behaviour is more or less the exact opposite of what was seen in Berwyn since these stations are relatively close, with only one station (Argyle) in between. However, the positive behind this is that the model is able to capture the seasonality of the ridership.

After looking at the prediction results for the three stations, it can be confirmed that the model is able to make predictions that are not totally random, and it has learned some patterns. However, it is unreliable, as can be seen in the performance disparity between the (relatively good) predictions of Damen and Berwyn and the (bad) predictive performance for Lawrence. At the same time, the loss is high even for this model, and it is overfitting the training data. The reason behind this predictive performance needs to be studied so that it can be improved.

4.2 Discussion and Future Work

A hypothesis that I have for this is that from the original data, the model can get some idea and find an underlying pattern, but it's not that good, and more features need to be provided to the model so that it has a more holistic view that can contribute to its understanding of the problem more. At the same time, the model was trained on ridership data for the last 5 years and maybe increasing the size of the training dataset could also help.
For the performance evaluation of the current model, just the value of RMSE loss was used as a metric. However, a more in-depth performance analysis is also required by using other features such as R^2, etc.

The most surprising part of this project was the fact that the model was able to even learn what it did, because, for most of the stations, the graph based network structure features were more or less similar (of course, with some exceptions), due to a large proportion of the stations having just two connections to the previous and the next stations, and as these for a large majority of the entire feature set. However, it could just be that the model's weights largely just ignored these features due to this very reason, and thus, the feature importance for these features would be fairly low.

4.2.1 Future Work

As stated before, the project was an implementation of Ding et. al., but for Chicago's L trains instead of the Shanghai metro. However, only a subset of techniques could be implemented from the original paper due to either the lack of enough data, compute or in the interest of time. For future work, I would like to start by implementing all the remaining techniques used in the paper:

- For features, the paper uses more variables/features than are used in the project. In addition to the current data, I would like to use other modes of transit, such as taxis, divvy bikes, etc. Additionally, for built environment features, in addition to population density and housing prices, other features, such as presence of major POIs such as schools, hospitals, tourist spaces etc around a station can also be included, however, to the best of my knowlegde, there isnt a directly available dataset for either of these, and using them, while possible, will be non-trivial.
- In addition to the above features, the paper also implements Graph attention, calculates graph attention features for each station based on the graph(s) and concatenates them with the above features. This graph attention model is trained alongside the predictor network. Having graph attention will surely improve the performance of the current model by a lot.
- This is not something that was done in the paper. Generally, people travel to and from a specific station, and that could give rise to pairs of stations, between which large groups of people often travel. For example, stations in the loop, such as Clark/Lake, can see a lot of inflow from multiple stations in the morning and outflow in the evening, say, for example, Irving Park, and these can be considered a pair. I would like to see if there is a way to find these pairs, and if adding these pairs as features can help improve the performance.

5 References

This section gives a list of sources that were used as a reference in this project. This will include the papers used/referred to, the tools used, as well as the AI help that was taken while writing code:

Research Papers

F. Ding, Y. Liang, Y. Wang, Y. Tang, Y. Zhou, and Z. Zhao, "A graph deep learning model for station ridership prediction in expanding metro networks," in Proc. 2nd ACM SIGSPATIAL Int. Workshop Advances Urban-AI, 2024, pp. 6–14, doi:10.1145/3681780.3697247.

Blogs

Spatial indexing in geospatial data to find out if something exists in a polygon: https://medium.com/@stacyfuende/python-spatial-indexing-accelerating-geographic-queries-f9bde82b8fe3

Tools Used

- Programming Language: Python
- Graphing tool: Networkx (Python Library)
- Library for training the Neural Network: Pytorch
- Library for parsing and using shapefiles: Shapely and Geopandas

AI Use: ChatGPT

1. Building the 'L' network as a graph without manually creating the adjacency matrix:

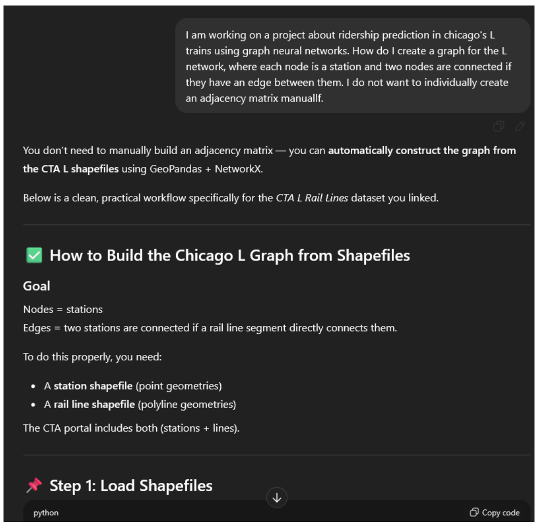


Figure 8: (i)

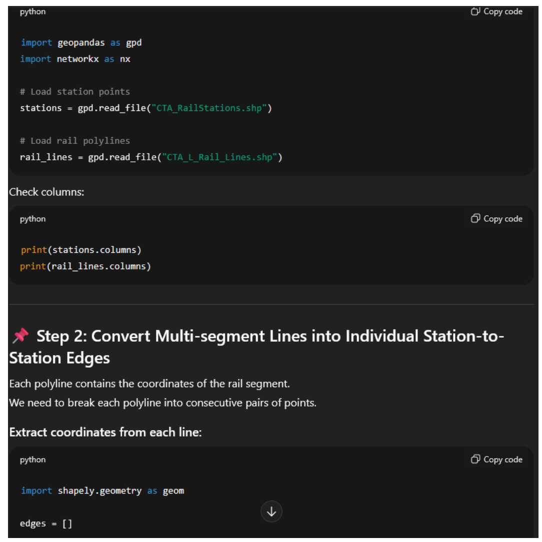


Figure 9: (ii)

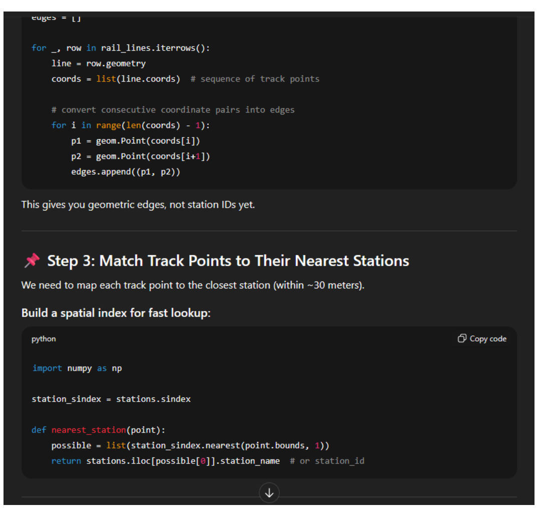


Figure 10: (iii)

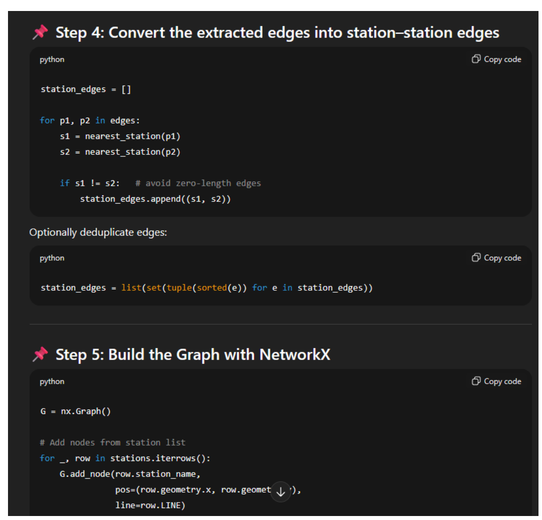
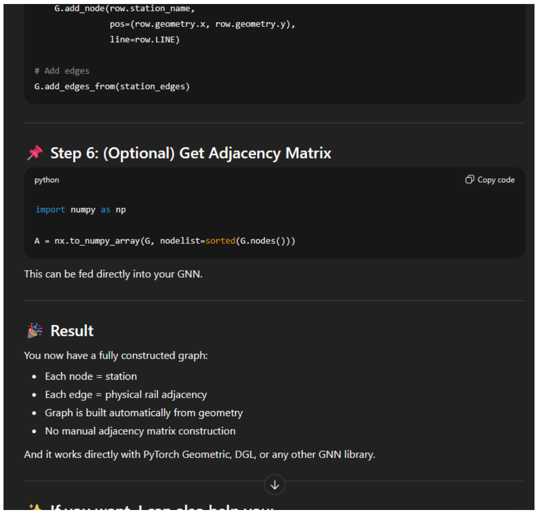


Figure 11: (iv)



```
    G.add_node(row.station_name,
               pos=(row.geometry.x, row.geometry.y),
               line=row.LINE)

# Add edges
G.add_edges_from(station_edges)
```

📌 Step 6: (Optional) Get Adjacency Matrix

python Copy code

```
import numpy as np

A = nx.to_numpy_array(G, nodelist=sorted(G.nodes()))
```

This can be fed directly into your GNN.

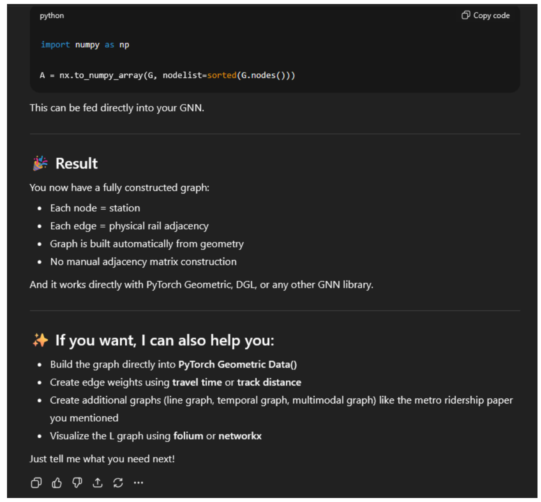
🎉 Result

You now have a fully constructed graph:

- Each node = station
- Each edge = physical rail adjacency
- Graph is built automatically from geometry
- No manual adjacency matrix construction

And it works directly with PyTorch Geometric, DGL, or any other GNN library.

Figure 12: (v)



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- No manual adjacency matrix construction

And it works directly with PyTorch Geometric, DGL, or any other GNN library.

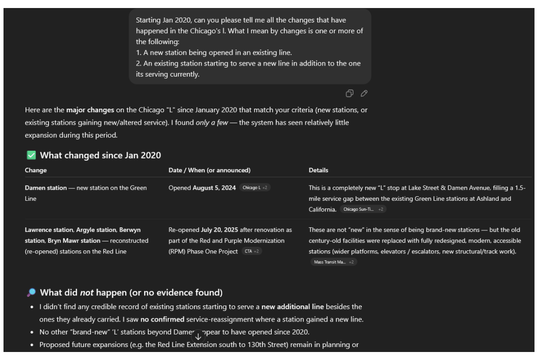
✨ If you want, I can also help you:

- Build the graph directly into **PyTorch Geometric Data()**
- Create edge weights using **travel time** or **track distance**
- Create additional graphs (line graph, temporal graph, multimodal graph) like the metro ridership paper you mentioned
- Visualize the L graph using **folium** or **networkx**

Just tell me what you need next!

Figure 13: (vi)

2. Asking about the changes in the CTA network from Jan 2020 till now



Starting Jan 2020, can you please tell me all the changes that have happened in the Chicago's l. What I mean by changes is one or more of the following:
1. A new station being opened in an existing line.
2. An existing station starting to serve a new line in addition to the one its serving currently.

Here are the **major changes** on the Chicago "L" since January 2020 that match your criteria (new stations, or existing stations gaining new/altered service). I found *only a few* — the system has seen relatively little expansion during this period.

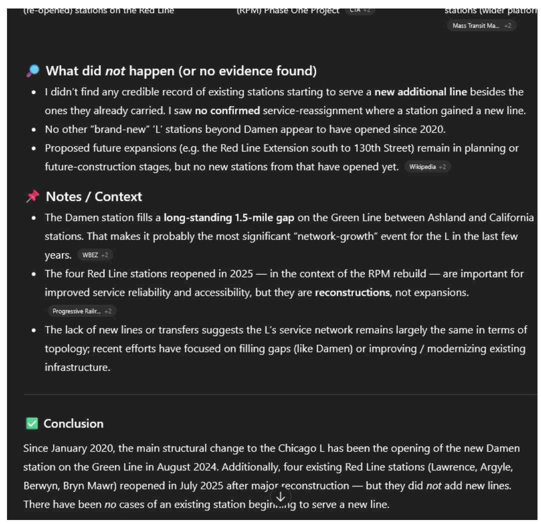
✅ What changed since Jan 2020

Change	Date / When (or announced)	Details
Damen station — new station on the Green Line	Opened **August 5, 2024** Chicago L +2	This is a completely new "L" stop at Lake Street & Damen Avenue, filling a 1.5-mile service gap between the existing Green Line stations at Ashland and California. Chicago Sun-Ti... +2
Lawrence station, Argyle station, Berwyn station, Bryn Mawr station — reconstructed (re-opened) stations on the Red Line	Re-opened **July 20, 2025** after renovation as part of the Red and Purple Modernization (RPM) Phase One Project CTA +2	These are not "new" in the sense of being brand-new stations — but the old century-old facilities were replaced with fully redesigned, modern, accessible stations (wider platforms, elevators / escalators, new structural/track work). Mass Transit Ma... +2

🔎 What did *not* happen (or no evidence found)

- I didn't find any credible record of existing stations starting to serve a **new additional line** besides the ones they already carried. I saw **no confirmed** service-reassignment where a station gained a new line.
- No other "brand-new" 'L' stations beyond Damen appear to have opened since 2020.
- Proposed future expansions (e.g. the Red Line Extension south to 130th Street) remain in planning or

Figure 14: (i)



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- No other "brand-new" 'L' stations beyond Damen appear to have opened since 2020.
- Proposed future expansions (e.g. the Red Line Extension south to 130th Street) remain in planning or future-construction stages, but no new stations from that have opened yet. Wikipedia +2

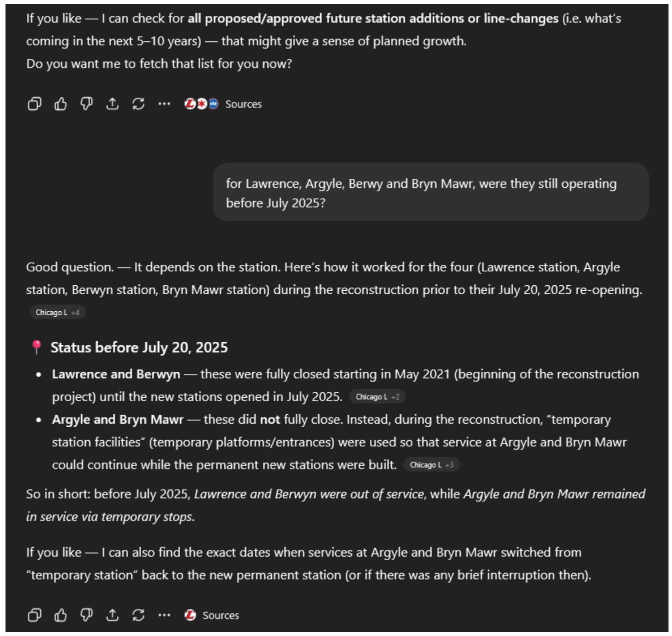
📌 Notes / Context

- The Damen station fills a **long-standing 1.5-mile gap** on the Green Line between Ashland and California stations. That makes it probably the most significant "network-growth" event for the L in the last few years. WBEZ +2
- The four Red Line stations reopened in 2025 — in the context of the RPM rebuild — are important for improved service reliability and accessibility, but they are **reconstructions**, not expansions. Progressive Railr... +2
- The lack of new lines or transfers suggests the L's service network remains largely the same in terms of topology; recent efforts have focused on filling gaps (like Damen) or improving / modernizing existing infrastructure.

✅ Conclusion

Since January 2020, the main structural change to the Chicago L has been the opening of the new Damen station on the Green Line in August 2024. Additionally, four existing Red Line stations (Lawrence, Argyle, Berwyn, Bryn Mawr) reopened in July 2025 after major reconstruction — but they did *not* add new lines. There have been *no* cases of an existing station beginning to serve a new line.

Figure 15: (ii)



If you like — I can check for **all proposed/approved future station additions or line-changes** (i.e. what's coming in the next 5–10 years) — that might give a sense of planned growth.
Do you want me to fetch that list for you now?

Sources

for Lawrence, Argyle, Berwy and Bryn Mawr, were they still operating before July 2025?

Good question. — It depends on the station. Here's how it worked for the four (Lawrence station, Argyle station, Berwyn station, Bryn Mawr station) during the reconstruction prior to their July 20, 2025 re-opening. Chicago L +4

📍 **Status before July 20, 2025**

- **Lawrence and Berwyn** — these were fully closed starting in May 2021 (beginning of the reconstruction project) until the new stations opened in July 2025. Chicago L +2
- **Argyle and Bryn Mawr** — these did **not** fully close. Instead, during the reconstruction, "temporary station facilities" (temporary platforms/entrances) were used so that service at Argyle and Bryn Mawr could continue while the permanent new stations were built. Chicago L +3

So in short: before July 2025, *Lawrence and Berwyn were out of service*, while *Argyle and Bryn Mawr remained in service via temporary stops.*

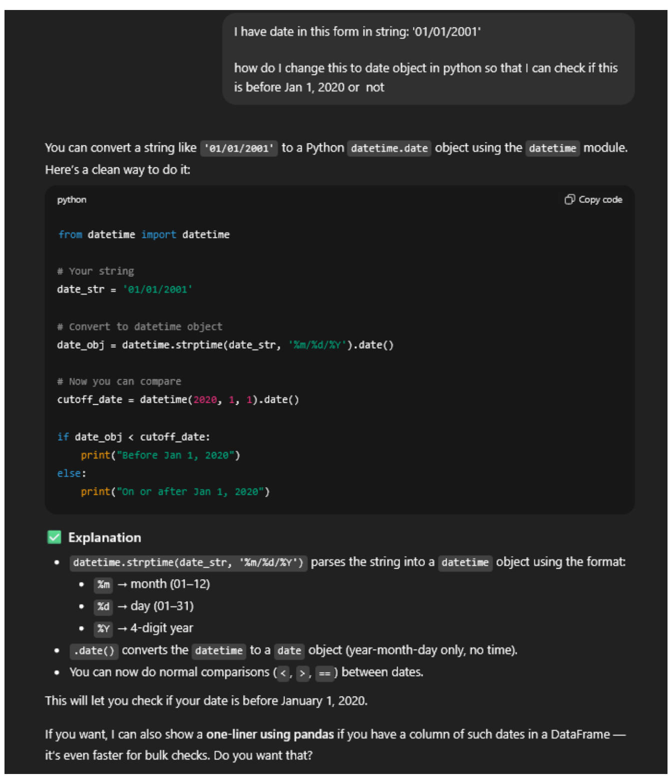
If you like — I can also find the exact dates when services at Argyle and Bryn Mawr switched from "temporary station" back to the new permanent station (or if there was any brief interruption then).

Sources

Figure 16: (iii)

Note: While I originally used ChatGPT to get a concise list in one place, all the information was verified to be correct before proceeding. The model did not say anything about the Racine station closure but this was then discovered in the ridership dataset.

3. How to convert a date from a string format to a datetime and check if it's less than a specific date:



I have date in this form in string: '01/01/2001'

how do I change this to date object in python so that I can check if this is before Jan 1, 2020 or not

You can convert a string like `'01/01/2001'` to a Python `datetime.date` object using the `datetime` module. Here's a clean way to do it:

python Copy code

```python
from datetime import datetime

# Your string
date_str = '01/01/2001'

# Convert to datetime object
date_obj = datetime.strptime(date_str, '%m/%d/%Y').date()

# Now you can compare
cutoff_date = datetime(2020, 1, 1).date()

if date_obj < cutoff_date:
    print("Before Jan 1, 2020")
else:
    print("On or after Jan 1, 2020")
```

✅ **Explanation**

- `datetime.strptime(date_str, '%m/%d/%Y')` parses the string into a `datetime` object using the format:
  - `%m` → month (01–12)
  - `%d` → day (01–31)
  - `%Y` → 4-digit year
- `.date()` converts the `datetime` to a `date` object (year-month-day only, no time).
- You can now do normal comparisons (`<`, `>`, `==`) between dates.

This will let you check if your date is before January 1, 2020.

If you want, I can also show a **one-liner using pandas** if you have a column of such dates in a DataFrame — it's even faster for bulk checks. Do you want that?

4. Creating the reference/citation link for the paper by Ding et. al.

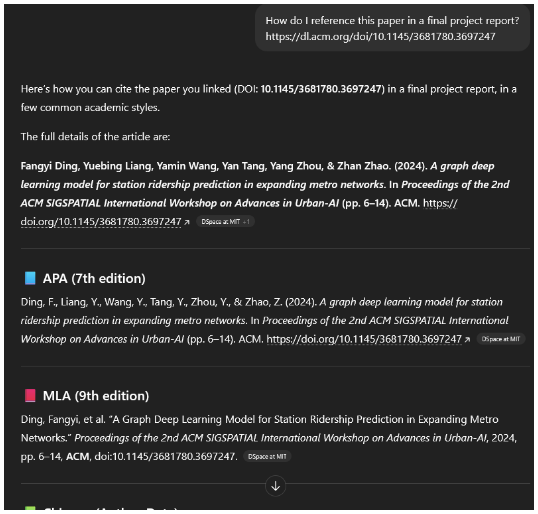


Figure 17: (i)

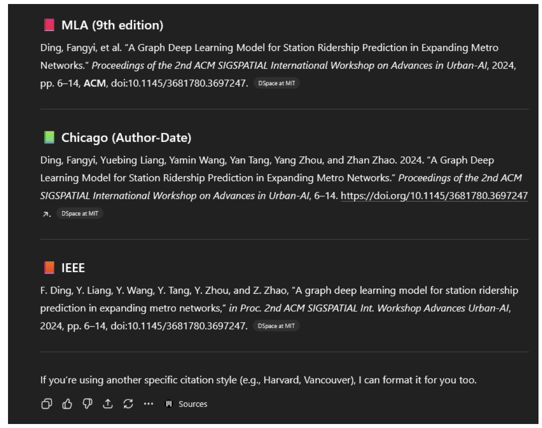


Figure 18: (ii)

6 Appendix

This is the appendix section that contains code and all the data sources used in the project. The code for the project is quite extensive and it would not be practical to put it all in this report. Same is the case for the data, the data sources are detailed in the report in section 2. Therefore, instead of pasting the code or its screenshots here, I am giving a link to the github repository which contains the code. The code can be found here: https://github.com/AmanSavaria1402/CS579_Proj.